

Unity requires physically enforced agreement

Sebastian Bobadilla-Suarez

Independent Researcher

`sebastian.bobadilla.s@gmail.com`

ORCID: 0000-0002-5951-0772

Abstract

Human conscious experience is unified: the brain's many local descriptions of a shared situation agree. We argue that unity requires this agreement to be physically enforced, produced and maintained by a coupling through which the graded physical state of each region continuously pulls the regions it overlaps toward agreement. Clocked digital hardware computes agreement instead. Its noise margins make every content readout locally constant in micro state, and we prove that a readout with a margin, under continuous micro dynamics, depends on no region's graded state; content read from bits is therefore never physically enforced, whatever algorithm the bits run. For a coupled sheet of oscillators we show that the width of the coupling kernel sets how fast agreement decays with distance, and that only a kernel tail slower than r^{-4} makes it independent of distance. The premise yields interventional tests on cortex and on artificial systems, and we state the outcomes that would refute it on each.

1 Introduction

Premise P (physical unity). A system’s contents are unified only if each region’s contents are read from graded content variables that depend continuously on the graded physical state of the regions it overlaps, through a coupling that contracts their disagreement.

P is a necessary condition for unity. It rejects functionalism about unity: a perfect simulation of a coupled field reproduces the field’s content-level dynamics and fails P. The paper makes three claims:

1. A content readout with a noise margin, under continuous micro dynamics, depends on the graded state of no region, so content read from bits is not physically enforced whatever algorithm the bits run (Section 4).
2. On a noisy coupled sheet, the width of the coupling kernel sets how fast agreement decays with distance, and only a kernel tail slower than r^{-4} makes agreement independent of distance (Section 5).
3. P predicts that a sub-threshold perturbation of one cortical region shifts the content decoded in an overlapping region toward agreement, and that no perturbation below a digital system’s noise margin shifts any of its contents (Section 8).

Every formal result the paper states is machine-checked in Lean 4, and Appendix A maps each one to its declaration.

2 Synchrony is not unity

Equal phases coexist with incompatible local descriptions, and compatible local descriptions determine a unique global state. Unity is therefore a property of what overlapping regions describe, and synchrony is at most a route to it.

3 Unity as agreement over a declared cover

Agreement is measured between regions that decode a shared variable. The cover of regions is fixed by decodability before agreement is measured, so that the measurement cannot be emptied by the choice of cover.

4 Physical enforcement versus routed agreement

A system is a micro dynamics f on the joint micro state $x = (x_i)_i$ of its regions, read by a content map π that assigns a content to each region. Its content level is closed when the next content is a function of the current one, $\pi \circ f = g \circ \pi$ for some content dynamics g . A readout has a noise margin on an operating set when every micro state in that set has a neighbourhood on which the readout is constant. Content depends gradedly on a region when no neighbourhood of that region's micro state leaves the next content unchanged, the others held fixed. The overlap discrepancy, a measure of disagreement between the contents of overlapping regions, contracts near a state when one step of f shrinks it by a factor below one throughout a neighbourhood of that state, so that perturbations below every threshold are pulled back toward agreement along with the rest. Agreement is physically enforced at a state when content depends gradedly on each region there and the overlap discrepancy contracts near it; P requires it at the states the system occupies while its contents are unified.

If the micro dynamics is continuous and carries a state into a margin's operating set, no region's micro state enters the next content gradedly. A thresholded continuous quantity has a margin off its threshold, and a finite word of such readouts has one too, so the conclusion holds for any content read from bits.

The theorem places the criterion one level below the content dynamics,

and the placement is forced. Contraction of disagreement at the content level is computable: a digital system running a consensus algorithm, or integrating a coupled field, has a content map whose successor contracts the disagreement between overlapping regions exactly as the field does. A criterion stated on the content dynamics would classify that system with the field. P is stated on the dependence of content on graded micro state, and there the two differ: the field's content moves with every graded change in the regions it overlaps, while the simulation's content is read from bits held inside their margins and moves with none of them.

This is the simulation objection, and P answers it by denial. A perfect simulation of a coupled cortical field reproduces the field's content-level trajectory and is not unified in the sense P requires, because the agreement in it is computed from contents rather than enforced by the graded state that carries them. That is the commitment P makes about unity, stated on the same footing as a workspace theory's identification of consciousness with broadcast: a premise whose consequences can be tested, which Section 8 does.

Categorical contents have margins in every substrate. The dominant percept in binocular rivalry, a perceived category, or any class selected by an argmax is constant on each side of its boundary, so away from the boundary the margin theorem denies it graded dependence in cortex exactly as in silicon. P is therefore stated on the graded content variables from which categories are read, such as a location, an orientation, a time or the relative strength of competing percepts, and a category meets P through the variable that carries it. A category read from a carrier depends gradedly on a region only where the carrier does. The restatement opens no route for a digital machine: every variable it computes, whether a number, a vector of numbers or a category, is a function of its bit word, and after any number of steps that keep its bits off their thresholds such a function depends gradedly on no region. The voltages beneath the bits carry none of the machine's contents,

at that step or any later one, so they are not a carrier.

5 How far a physical coupling enforces agreement

At identical natural frequencies a noisy coupled sheet is a gradient system. In the harmonic approximation the variance of the phase difference between two sites is the noise strength times their effective resistance, with the coupling constants as conductances. Under short-range coupling that resistance grows as the logarithm of distance, with a coefficient inversely proportional to the kernel's second moment; a kernel tail slower than r^{-4} bounds it. The logarithm is the Gaussian form of the absence of true long-range order in two dimensions under short-range coupling (Mermin & Wagner, 1966), and the plane-rotator sheet has a Kosterlitz–Thouless transition (Fröhlich & Spencer, 1981), the order–disorder crossover that the simulations below locate. Heterogeneous natural frequencies make short-range coupling weaker still: phases then desynchronize in every dimension up to four (Hong et al., 2005).

The logarithmic growth is a lower bound for every kernel of bounded range. Group the sites into shells around a site a , so that the coupling joins only sites in the same or adjacent shells. The shells then act as conductors in series, and the effective conductance from a to any site beyond shell n is at most $(\sum_{k < n} 1/C_k)^{-1}$, where C_k is the coupling crossing from shell k to shell $k + 1$. On a planar sheet the perimeter of shell k grows linearly in k , so $C_k \leq c(k + 1)$ for a constant c set by the kernel's strength and width, and the effective resistance is at least $\log(n + 1)/c$. Widening the kernel enlarges its shells and raises c , which slows the growth and leaves it unbounded. Distance-independent agreement on a sheet therefore needs coupling of unbounded range.

A tail slower than r^{-4} bounds the resistance at every distance. Run a

chain of site sets out from a through discs of radius 2^j and back in to b , the two widest discs spanning the separation. The phase difference across a and b telescopes through the discs' mean phases, and the drop between the means of two consecutive discs is bounded by the coupling energy between them, shared among the pairs of their sites. A disc of radius 2^j on a planar sheet holds at least $\alpha 4^j$ sites, and a kernel $K(r) \gtrsim r^{-(2+\sigma)}$ couples two sites of consecutive discs by at least $\beta 2^{-(2+\sigma)j}$. The number of pairs grows as 16^j and their coupling falls as $2^{-(2+\sigma)j}$, so the drops shrink geometrically with ratio $q = 2^{-(2-\sigma)/2}$, which is below one for $\sigma < 2$, and the effective conductance is at least $(1 - q)^2 \alpha^2 \beta / 8$ however far apart a and b lie.

Simulated sheets at identical frequencies, of side 64 and 128, with nearest-neighbour, exponential and power-law kernels of tail exponent $\sigma = 1$ and $\sigma = 3$ at a common row sum, follow the harmonic identity in the ordered phase. The measured phase variance is at most 1.15 times $D R_{\text{eff}}$ at $D = 0.1$ and at most 1.26 times at $D = 0.4$, the excess rising with noise as the cosine coupling softens, and a logarithm fits its growth with distance better than a line in 16 of the 16 runs at those noise levels. At $D = 1.2$ the nearest-neighbour sheet disorders: the mean chord discrepancy $\langle 1 - \cos(\theta_a - \theta_b) \rangle$ at the largest separation is 1.00. The exponential and $\sigma = 1$ sheets stay ordered at that noise (0.46 and 0.50), and the $\sigma = 3$ sheet nearly disorders (0.92), so the kernel's width sets the order-disorder crossover as well as the rate of decay. Exact resistance sums at sheet sides 128, 512 and 1024 separate width from tail. The exponential kernel, with 25 times the nearest-neighbour stiffness, gives resistance at half the side of 0.531, 0.549 and 0.558, still growing; the $\sigma = 1$ tail gives 0.580, 0.577 and 0.577, and the $\sigma = 3$ tail, steeper than r^{-4} , gives 1.198, 1.448 and 1.572, growing as a short-range kernel does. Over a finite cortical extent a wide short-range kernel and a slow tail therefore differ by a slowly growing logarithm, and the measurement that separates them needs the kernel's second moment as well as its tail exponent.

Three facts hold without the harmonic approximation. Adding coupling

never raises the effective resistance, so a long-range tail added to a short-range kernel can only improve agreement, whatever its shape. A direct coupling between two sites caps their resistance at its reciprocal, at any distance. A Lipschitz encoder carries phase agreement to content agreement: the mean-square content discrepancy is at most the squared Lipschitz constant times the mean-square phase difference. With the harmonic identity, an encoder of Lipschitz constant L at noise strength D therefore has root-mean-square content discrepancy at most $L\sqrt{D R_{\text{eff}}}$ between two sites of effective resistance R_{eff} .

Content can agree where phases differ. An encoder that reads the phases of a stencil around a site relative to the phase at that site reports the same content at every site of a plane wave, whatever its gradient, and at every site of a winding on a ring or torus. For a wave plus a departure, the content discrepancy between two sites of a Lipschitz relative encoder is at most its Lipschitz constant times the departure's relative readings at the two sites, with the gradient of the wave and the separation of the sites absent from the bound. Travelling waves and windings, the configurations that attain the worst case of a hop-by-hop chain bound, thus cost a relative encoder nothing; what it pays for is the local roughness of the field.

6 The unity window

Agreement cannot arrive before the causal cone of the coupling allows, and disagreement decays at the relaxation rate of the coupling. Unity at a given time scale requires the interval between the two to be nonempty.

The lower edge is causal. A report at one site that tracks a variable at another to within ε , for two values of the variable more than 2ε apart, is possible after T rounds of the coupling only if the second site lies in the first site's causal past for T rounds. The upper edge is relaxation. If the coupling contracts the overlap discrepancy by a factor $\kappa < 1$ at every step on a set

of states it keeps, a discrepancy starting at most d_0 is at most ε from step $\log(d_0/\varepsilon)/\log(1/\kappa)$ on. The estimate is sharp in form: a discrepancy that shrinks by at most a factor ρ per step is still above ε at every earlier step than $\log(d_0/\varepsilon)/\log(1/\rho)$. The window opens at the later of the causal deadline and the relaxation time, and unity at the time scale of a content requires it to open before that content changes.

In cortex the longest cortico-cortical fibres run about 15 cm (Nunez & Srinivasan, 2014), callosal axons conduct at median velocities of 4.9 to 8.8 m/s and the fastest of them at up to 20 m/s (Caminiti et al., 2009), and a chemical synapse adds about 0.5 ms (Katz & Miledi, 1965; Sabatini & Regehr, 1996). The synaptic causal deadline across the cortex is therefore 8 ms along the fastest axons, and typical synaptic arrival takes 18 to 31 ms. The unconscious integration that precedes a conscious percept lasts up to 400 ms (Herzog et al., 2016). Reading agreement as a 10-fold reduction of disagreement, decaying exponentially, the window at that time scale is nonempty when the coupling’s relaxation time is at most 160 ms with typical arrival along the slower fibres and 166 ms along the faster. The causal deadline thus takes a small part of the content’s time scale, and whether the window opens turns on the relaxation time.

7 Current artificial systems

A transformer running on clocked digital hardware reads every content from bits held within their noise margins, so the result of Section 4 applies to it. Hardware whose contents are graded physical quantities coupled without barriers, such as analog, in-memory or neuromorphic substrates, can satisfy P.

The criterion concerns how a substrate is organised and says nothing of what it is made of. An analog crossbar whose column currents carry content and settle through shared conductances depends gradedly on every input

voltage, and an array of coupled electronic or optical oscillators contracts phase disagreement through a physical coupling exactly as a cortical sheet does. A neuromorphic chip passes where its contents are read from membrane potentials coupled continuously, and fails where they are read from spike counts latched at a clock edge. Biological tissue passes or fails on the same terms: a neural circuit whose contents were read only from all-or-none events latched far from threshold would fall under the margin theorem as any digital circuit does.

Spikes are thresholded events, and what makes them graded carriers is their timing. The time at which a membrane potential rising through threshold crosses it moves continuously with the sub-threshold input, and postsynaptic potentials integrate spike times without sampling them, so spike timing carries graded dependence from one region to the next. A clocked latch compares the time of a transition with the clock edge, which is a threshold on the transition time, so the margin theorem applies to it with that time as the thresholded quantity. The clock is a noise margin in time, and it removes from a digital circuit exactly the graded timing that spikes keep.

Real digital hardware leaks: supply noise, crosstalk and temperature move the analog quantities behind every bit. The margin theorem locates where such leakage can reach content. If a content read as a word of thresholded continuous quantities depends gradedly on any region, some bit of the next state sits exactly on its threshold, and correct operation keeps every state off that set. The margin also has a width. If every bit of the next state is at least δ from its threshold, each bit is read from an L_v -Lipschitz quantity and the micro dynamics is L -Lipschitz, a change to one region smaller than $\delta/(LL_v)$ changes no bit. Leakage below that scale carries no content; leakage above it flips a bit, which is a logical error. Digital design places ordinary noise well below that scale: a textbook inverter's static noise margin is 41% of its supply voltage, against an expected supply variation of 10% (Rabaey et al., 2003), so the expected variation is 0.24 of the margin. The nondetermin-

ism of floating-point reduction order is a choice among logical trajectories and is outside both theorems: it varies which content is computed, not how content depends on graded state.

8 Tests

Graded cross-region dependence in cortex. Weak stimulation of one region below the behavioural threshold should shift the content decoded in an overlapping region, gradedly with stimulation strength and toward agreement with the stimulated region. Transcranial alternating current and weak intracortical fields supply the perturbation, and the decoded content is a graded variable, such as a location, an orientation or the strength of evidence for a percept, read from population activity in the overlapping region; a decoded category is locally constant away from its boundaries in any substrate and would stay unchanged whether P holds or not. An unchanged decoded content in the overlapping region, at stimulation strengths that measurably move the stimulated one, counts against P, since P then denies unity to the system whose unity motivates it.

Agreement as an attractor on both substrates. Perturb one region and measure whether agreement with its overlap partners is restored, and at what rate. In cortex the perturbation is stimulation and the measure is the disagreement between contents decoded from the perturbed region and its partners; P predicts restoration at the relaxation rate of Section 6, with a restoring pull proportional to the size of the perturbation below as well as above any threshold. In an artificial system the perturbation is activation patching of one layer, head or position, applied at two scales: below the scale at which the logical trajectory changes, and above it. A digital system can restore agreement above that scale, by computation; below it the margin theorem predicts that nothing moves at all. The outcome that counts against

the thesis on the artificial side is a graded shift of decoded content in one region, toward agreement, driven by an analog perturbation of another that stays inside every bit’s margin, such as a supply or timing perturbation below the error threshold.

The field’s causal cone. At the frequencies of neural activity the extracellular field is quasi-static (Gratny et al., 2017): it reaches every point of the tissue it spans with no propagation delay, while synaptic influence between distant regions arrives only after axonal conduction and synaptic transmission, no sooner than 8 ms across the longest cortico-cortical fibres. Re-agreement between two regions after a local perturbation, faster than the synaptic route allows, would identify a non-synaptic contribution to the coupling and measure the reach of the kernel tail of Section 5 directly.

The strength of the field sets the separation at which the test decides anything. Endogenous fields in cortex peak near 2.4 mV/mm in vivo (Fröhlich & McCormick, 2010), networks of neurons detect fields from 0.14 mV/mm (Francis et al., 2003), and the potential of a local cortical source decays as $r^{-2.1}$ across millimetres (Rebollo et al., 2021). With the peak placed at the nearest distance of that measurement, the field of a local source falls to the detection threshold within 3.7 mm and misses it by 5 orders of magnitude at 15 cm. The field’s cone is therefore tested at millimetre separations, where activity has been observed to synchronize across a cut through cortex by the field alone (Rebollo et al., 2021); at centimetre separations P predicts that re-agreement arrives through synaptic coupling, which is graded as well. The timing is measurable with intracortical recording. The stimulation artefact saturates clinical amplifiers during the first milliseconds, masking the earliest monosynaptic responses (Keller et al., 2014), and early responses at distant contacts carry a volume-conducted component that scales with distance as the artefact does (Prime et al., 2020). Template subtraction of the artefact (Trebaul et al., 2016), multi-unit activity or current-source density at the dis-

tant site, which the passive spread of the local field potential does not reach (Kajikawa & Schroeder, 2011), and a deadline set for each subject from the tractography length of the connecting fibres (Caminiti et al., 2013) separate active re-agreement from the passive field that carries the stimulus.

Separating coupling from information. In every manipulation of cortex available so far, graded coupling and information flow move together: anaesthesia and callosal section remove both. A test that confirms P against functionalism must move one with the other held fixed. Coherence between cortical regions arises largely as a consequence of communication between them (Schneider et al., 2021), and transcranial alternating current applied to two regions drives both from a common source: where in-phase stimulation raised their synchrony it raised directed information flow with it (Reinhart & Nguyen, 2019). The coupling has to be supplied instead. Two regions whose natural connection is interrupted are bridged by stimulating each in proportion to the activity recorded at the other, through one of two loops: an analog loop, whose output moves with every graded change in its input, or a quantized loop, whose output is read from bits and so has a margin. Noise added to the analog loop matches the transfer entropy it carries, estimated from the recorded activity, to the quantized loop's. The two loops then carry the same information, and only the analog one couples the regions gradedly. Unity measured as in the next paragraph that follows the analog loop at matched transfer entropy supports P over functionalism, and unity that is the same under both loops counts against P. Intracortical stimulation in animals supports the design with decoded agreement as the measure; reported unity needs a bridged human preparation.

What counts as evidence about the phenomenal field. Every test above measures decoded agreement, which is agreement of the contents available to readout. If disunity of readout could always be set aside as disunity of access alone, no observation would bear on P. The observations that count as

evidence about the phenomenal field are therefore fixed in advance: responses that integrate stimuli across the regions in question when the task requires it, and reports of a single field of experience spanning them. A dissociation in which such responses and reports split while graded coupling between the regions is intact, or stay integrated after graded coupling between them is removed, is accepted as evidence against P.

Split brain. Callosal section removes the main route of both information and graded coupling between the hemispheres. Patients respond to stimuli in the two visual fields as if perception were divided, yet can report the presence of stimuli in either field with either hand, which has been read as divided perception with undivided consciousness (Pinto et al., 2017); whether the phenomenal field divides remains contested (de Haan et al., 2020), and a switch model, on which consciousness alternates between the hemispheres, is a live rival (Bayne, 2010). If the field stays undivided, unity survives on the residual subcortical routes. Those routes are synaptic and carry information too, so the case bears on P only if unity is preserved out of proportion to the information they carry, and it bears on graded coupling of any kind: synaptic coupling is graded through postsynaptic potentials and sub-threshold shifts in spike timing, and the extracellular field is one realization of it among several.

9 Related work

The closest formal neighbour is the account of computational closure of Rosas et al. (2024), on which a macroscopic level that is informationally, causally and computationally closed is the mark of emergent software in a physical system. The closure is the same property that Section 4 places at the content level of a digital system, and P reads it the other way: a level closed off from the graded state beneath it is one whose agreement is computed,

not enforced. Kleiner & Ludwig (2024) reach a related exclusion from a different premise: if the physical dynamics of a system is relevant to its consciousness, processors designed and verified to suppress deviations from their computational dynamics cannot be conscious. The margin theorem gives the suppression a formal statement, and P confines the premise to unity and ties it to a coupling that contracts disagreement, which is what makes it testable on both substrates. Seth (2026) argues that consciousness may depend on biological properties that computation does not capture; P locates the relevant property in the organisation of the substrate and admits non-biological hardware that has it (Section 7). Mortal computation, whose parameters are useful only on the hardware instance that learned them (Hinton, 2022; Ororbia & Friston, 2023), shares P’s refusal to separate the computation from the medium. Integrated information theory also denies experience to digital computers running faithful simulations of a brain (Tononi & Koch, 2015; Findlay et al., 2024); it does so through the causal structure of the hardware at its logical grain, while P turns on the dependence of content on graded physical state and on the contraction of disagreement. The electromagnetic field theory of McFadden (2020) identifies consciousness with the brain’s field, supported by evidence that endogenous fields entrain cortical activity (Fröhlich & McCormick, 2010); P takes the field as one realization of graded coupling among several.

10 Limitations

P is a premise and is not derived from the dynamics; the formal results show what follows from it and not that it holds. It is necessary for unity and not sufficient for consciousness: coupled pendulums satisfy it. The margin theorem concerns idealised digital readouts, and graded leakage in real hardware that carries content-relevant agreement would narrow its reach. The identification of phase variance with effective resistance rests on the har-

monic approximation at identical natural frequencies. The logarithmic lower bound on resistance is proved for kernels of bounded range, and the bound at every distance under a tail slower than r^{-4} is proved for a sheet that supplies the site counts and couplings of a dyadic chain of discs; that the logarithm is attained, and that a tail steeper than r^{-4} leaves the resistance growing, are established numerically on finite sheets. The unity window's edges are proved for a given contraction factor; the identification of that factor with the coupling's spectral gap is linear response. The relaxation time of cortical coupling is unmeasured, so the window estimate states the condition for the window to open at cortical time scales and leaves open whether cortex meets it; it also reads agreement as a declared reduction factor of disagreement. The step from phase agreement to content agreement assumes one oscillator population per content patch, which is a modelling choice about cortex. A long-range tail improves agreement only in proportion to its conductance against the short-range part, and the field of a local cortical source falls below the detection threshold within millimetres, by an estimate that places the field's peak at a declared distance; across centimetres graded coupling is carried chiefly by synapses, and the field's contribution there is a quantity to be measured. The dissociation of coupling from information has, so far, no human preparation in which reported unity can be measured across a bridged connection. Above the order-disorder crossover of Section 5 the harmonic identity fails and agreement is lost at every distance. Human data can falsify P but cannot by themselves confirm it against functionalism, because graded coupling and information flow co-vary in every manipulation available so far.

References

- Bayne, T. (2010). *The Unity of Consciousness*. Oxford University Press.
doi:10.1093/acprof:oso/9780199215386.001.0001.
- Caminiti, R., Ghaziri, H., Galuske, R., Hof, P. R., & Innocenti, G. M. (2009).

- Evolution amplified processing with temporally dispersed slow neuronal connectivity in primates. *Proceedings of the National Academy of Sciences*, 106(46), 19551–19556. doi:10.1073/pnas.0907655106.
- Caminiti, R., Carducci, F., Piervincenzi, C., Battaglia-Mayer, A., Confalone, G., Visco-Comandini, F., Pantano, P., & Innocenti, G. M. (2013). Diameter, length, speed, and conduction delay of callosal axons in macaque monkeys and humans: Comparing data from histology and magnetic resonance imaging diffusion tractography. *Journal of Neuroscience*, 33(36), 14501–14511. doi:10.1523/JNEUROSCI.0761-13.2013.
- de Haan, E. H. F., Corballis, P. M., Hillyard, S. A., Marzi, C. A., Seth, A., Lamme, V. A. F., Volz, L., Fabri, M., Schechter, E., Bayne, T., Corballis, M. C., & Pinto, Y. (2020). Split-brain: What we know now and why this is important for understanding consciousness. *Neuropsychology Review*, 30(2), 224–233. doi:10.1007/s11065-020-09439-3.
- Findlay, G., Marshall, W., Albantakis, L., David, I., Mayner, W. G. P., Koch, C., & Tononi, G. (2024). Dissociating artificial intelligence from artificial consciousness. *arXiv:2412.04571*. doi:10.48550/arXiv.2412.04571.
- Francis, J. T., Gluckman, B. J., & Schiff, S. J. (2003). Sensitivity of neurons to weak electric fields. *Journal of Neuroscience*, 23(19), 7255–7261. doi:10.1523/JNEUROSCI.23-19-07255.2003.
- Fröhlich, J., & Spencer, T. (1981). The Kosterlitz–Thouless transition in two-dimensional Abelian spin systems and the Coulomb gas. *Communications in Mathematical Physics*, 81(4), 527–602. doi:10.1007/BF01208273.
- Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may guide neocortical network activity. *Neuron*, 67(1), 129–143.
- Gratiy, S. L., Hanes, G., Denman, D., Hawrylycz, M. J., Koch, C., Einevoll, G. T., & Anastassiou, C. A. (2017). From Maxwell’s equations to the the-

- ory of current-source density analysis. *European Journal of Neuroscience*, 45(8), 1013–1023. doi:10.1111/ejn.13534.
- Herzog, M. H., Kammer, T., & Scharnowski, F. (2016). Time slices: What is the duration of a percept? *PLoS Biology*, 14(4), e1002433. doi:10.1371/journal.pbio.1002433.
- Hinton, G. (2022). The forward-forward algorithm: Some preliminary investigations. *arXiv:2212.13345*. doi:10.48550/arXiv.2212.13345.
- Hong, H., Park, H., & Choi, M. Y. (2005). Collective synchronization in spatially extended systems of coupled oscillators with random frequencies. *Physical Review E*, 72(3), 036217. doi:10.1103/PhysRevE.72.036217.
- Kajikawa, Y., & Schroeder, C. E. (2011). How local is the local field potential? *Neuron*, 72(5), 847–858. doi:10.1016/j.neuron.2011.09.029.
- Katz, B., & Miledi, R. (1965). The measurement of synaptic delay, and the time course of acetylcholine release at the neuromuscular junction. *Proceedings of the Royal Society B*, 161(985), 483–495. doi:10.1098/rspb.1965.0016.
- Keller, C. J., Honey, C. J., Mégevand, P., Entz, L., Ulbert, I., & Mehta, A. D. (2014). Mapping human brain networks with cortico-cortical evoked potentials. *Philosophical Transactions of the Royal Society B*, 369(1653), 20130528. doi:10.1098/rstb.2013.0528.
- Kleiner, J., & Ludwig, T. (2024). The case for neurons: A no-go theorem for consciousness on a chip. *Neuroscience of Consciousness*, 2024(1), niae037. doi:10.1093/nc/niae037.
- McFadden, J. (2020). Integrating information in the brain’s EM field: the cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1), niaa016.

- Mermin, N. D., & Wagner, H. (1966). Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic Heisenberg models. *Physical Review Letters*, 17(22), 1133–1136. doi:10.1103/PhysRevLett.17.1133.
- Nunez, P. L., & Srinivasan, R. (2014). Neocortical dynamics due to axon propagation delays in cortico-cortical fibers: EEG traveling and standing waves with implications for top-down influences on local networks and white matter disease. *Brain Research*, 1542, 138–166. doi:10.1016/j.brainres.2013.10.036.
- Ororbia, A., & Friston, K. (2023). Mortal computation: A foundation for biomimetic intelligence. *arXiv:2311.09589*. doi:10.48550/arXiv.2311.09589.
- Pinto, Y., Neville, D. A., Otten, M., Corballis, P. M., Lamme, V. A. F., de Haan, E. H. F., Foschi, N., & Fabri, M. (2017). Split brain: Divided perception but undivided consciousness. *Brain*, 140(5), 1231–1237. doi:10.1093/brain/aww358.
- Prime, D., Woolfe, M., O’Keefe, S., Rowlands, D., & Dionisio, S. (2020). Quantifying volume conducted potential using stimulation artefact in cortico-cortical evoked potentials. *Journal of Neuroscience Methods*, 337, 108639. doi:10.1016/j.jneumeth.2020.108639.
- Rabaey, J. M., Chandrakasan, A., & Nikolić, B. (2003). *Digital Integrated Circuits: A Design Perspective* (2nd ed.). Prentice Hall.
- Rebollo, B., Telenczuk, B., Navarro-Guzman, A., Destexhe, A., & Sanchez-Vives, M. V. (2021). Modulation of intercolumnar synchronization by endogenous electric fields in cerebral cortex. *Science Advances*, 7(10), eabc7772. doi:10.1126/sciadv.abc7772.
- Reinhart, R. M. G., & Nguyen, J. A. (2019). Working memory revived

- in older adults by synchronizing rhythmic brain circuits. *Nature Neuroscience*, 22(5), 820–827. doi:10.1038/s41593-019-0371-x.
- Rosas, F. E., Geiger, B. C., Luppi, A. I., Seth, A. K., Polani, D., Gastpar, M., & Mediano, P. A. M. (2024). Software in the natural world: A computational approach to hierarchical emergence. *arXiv:2402.09090*. doi:10.48550/arXiv.2402.09090.
- Sabatini, B. L., & Regehr, W. G. (1996). Timing of neurotransmission at fast synapses in the mammalian brain. *Nature*, 384(6605), 170–172. doi:10.1038/384170a0.
- Schneider, M., Brogini, A. C., Dann, B., Tzanou, A., Uran, C., Sheshadri, S., Scherberger, H., & Vinck, M. (2021). A mechanism for inter-areal coherence through communication based on connectivity and oscillatory power. *Neuron*, 109(24), 4050–4067. doi:10.1016/j.neuron.2021.09.037.
- Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism. *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- Tononi, G., & Koch, C. (2015). Consciousness: here, there and everywhere? *Philosophical Transactions of the Royal Society B*, 370(1668), 20140167. doi:10.1098/rstb.2014.0167.
- Trebaul, L., Rudrauf, D., Job, A.-S., Măliia, M. D., Popa, I., Barborica, A., Minotti, L., Mîndruță, I., Kahane, P., & David, O. (2016). Stimulation artifact correction method for estimation of early cortico-cortical evoked potentials. *Journal of Neuroscience Methods*, 264, 94–102. doi:10.1016/j.jneumeth.2016.03.002.

A Formal results

Each formal result the paper states is a theorem of the Lean 4 development in the project repository, in its four `Phase10` modules. Table 1 maps each result to its declaration and to the section that argues from it. Every theorem rests only on Lean’s foundational axioms `propext`, `Classical.choice` and `Quot.sound`, which the build verifies for every declaration of the development. Small concrete models in `Examples/` meet the hypotheses of the bounds and show where they are attained.

Table 1: The paper’s formal results and the Lean declarations that carry them.

Result	Section	Declaration
A readout with a margin, under continuous micro dynamics, has no graded dependence on any region	4	<code>not_physicallyEnforced_of_margin</code>
The same for any content read from a finite word of thresholded continuous quantities	4	<code>not_physicallyEnforced_of_bits</code>
A thresholded continuous quantity has a margin off its threshold	4	<code>hasMargin_threshold</code>
A category read from a carrier depends gradedly on a region only where the carrier does	4	<code>GradedDependence.of_comp</code>
Every function of a bit word, after steps that keep the bits off their thresholds, depends gradedly on no region	4	<code>not_gradedDependence_iterate_of_bits</code>
Shells in series bound the effective conductance	5	<code>conductance_le_shell</code>
Crossing coupling at most $c(k + 1)$ gives resistance at least $\log(n + 1)/c$	5	<code>conductance_mul_log_le</code>
An averaging chain from a to b bounds the resistance by $2(\sum_k \rho_k)^2$	5	<code>one_le_conductance_of_chain</code>
A tail $K(r) \gtrsim r^{-(2+\sigma)}$ with $\sigma < 2$ bounds the conductance below at every distance	5	<code>le_conductance_of_powerLaw</code>
Adding coupling never raises the effective resistance	5	<code>conductance_mono</code>
A direct coupling caps the resistance at its reciprocal	5	<code>le_conductance_of_edge</code>
A Lipschitz encoder carries mean-square phase agreement to content agreement	5	<code>integral_dist_sq_le</code>
A relative encoder reports one content at every site of a wave	5	<code>relContent_eq_of_wave</code>
A relative encoder’s discrepancy is bounded by the departure from a wave	5	<code>dist_relContent_le</code>
Tracking a remote variable requires it to lie in the causal past	6	<code>mem_ball_of_reconstructs</code>
Contraction by κ reaches ε by step $\log(d_0/\varepsilon)/\log(1/\kappa)$	6	<code>disc_iterate_le_of_log_le</code>
A discrepancy shrinking by at ₂₁ most ρ per step is above ε before $\log(d_0/\varepsilon)/\log(1/\rho)$	6	<code>lt_disc_iterate_of_lt_log</code>
Graded dependence of a bit word requires some bit exactly on its threshold	7	<code>exists_threshold_of_gradedDependence</code>
A one-region change below $\delta/(LL_v)$ changes no bit	7	<code>bits.eq.of_lipschitz</code>